

Barrier, Binary and Asian Option Pricing

Miura Financial Research
Quantitative Research Team

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Abstract

In these notes, we present a concise overview of several well-established methods for pricing barrier, Asian, and binary options. Specifically, we explore the Binomial, Trinomial, Black-Scholes, and Monte Carlo methods. Following a brief explanation of each approach, we apply these techniques using real data from the S&P 500 index. Additionally, for the computation of the volatility parameter (σ) in all of these models, we employ a range of historical volatility estimators, including those introduced by Parkinson, Garman and Klass, Rogers and Satchell, and Yang and Zhang.

Introduction

The pricing of exotic options, particularly those that are path-dependent, represents a crucial area of research in empirical finance due to their distinctive features. Although these options are traded less frequently than standard options, they hold significant importance in financial markets, particularly for risk management. While standard options, especially American options, dominate market activity, quantifying the role of exotic options presents substantial challenges. This challenge primarily stems from the fact that most exotic options are predominantly traded in over-the-counter (OTC) markets, which makes it difficult to obtain accurate data on trading volumes, open interest, and other pertinent trading metrics.

Despite their relatively niche nature, exotic options are central to quantitative finance, largely due to the complexity of their payoff structures. The mathematical models used to value these options tend to be more advanced than those for standard options, rendering their study both challenging and intriguing. In these notes, we provide a brief overview of several key methods employed in the pricing of barrier, Asian, and binary options.

Given the wide array of available models, we focus on a select few core approaches: analytical formulas, particularly within the Black-Scholes framework; Monte Carlo simulations; and lattice models, specifically binomial and trinomial trees. Special attention is given to the challenges encountered when pricing barrier options using Monte Carlo and lattice methods. We also discuss some of the main techniques proposed to address these challenges. For Monte Carlo simulations, we examine the solution suggested by Baldi, Caramellino, and Iovino [1]. In the context of binomial and trinomial lattices, we consider the adjustments proposed by Derman et al. [15], Kamrad and Ritchken [23], and Ritchken [30].

The final section of these notes presents a straightforward empirical application, utilizing real data from the S&P 500 index. For model calibration, we rely on the par yield curve derived from the U.S. government bond market. Furthermore, we incorporate various volatility estimators, including the close-to-close volatility estimator, as well as those based on OHLC data, such as the estimators proposed by Parkinson [28], Garman and Klass [18], Rogers and Satchell [34], and Yang and Zhang [36]. It is important to note that the calibration methodology presented here does not fully reflect the common practices of market practitioners. In particular, implied volatility is typically employed in practice, and Overnight Index Swaps (OIS) are generally used for determining the risk-free rate.

The primary objective of these notes is to provide clear and concise explanations of the models employed in the pricing of these types of options.

1 Options

Options Market: characteristics and recent developments

Options have established themselves as one of the most versatile and essential instruments in global financial markets. Widely utilized for hedging, risk management, portfolio diversification, and speculative trading, options offer a unique combination of flexibility and risk-reward dynamics.

The accessibility of options has attracted a broad range of participants, from institutional investors seeking efficient risk management tools to retail traders exploring short-term speculative opportunities. This diverse participation underpins the critical role of options in modern financial systems.

In the 2023 report by FIA [1], the global options market is predominantly dominated by the United States, which hosts several major exchanges, including the Chicago Board Options Exchange (CBOE), the NASDAQ Options Market, and the New York Stock Exchange (NYSE). Specialized platforms, such as MIAX and BOX, further bolster the U.S.'s role as the central hub of global options trading. Together, these exchanges process a substantial portion of global trading volumes, benefiting from advanced technology and substantial liquidity pools.

Beyond the United States, the Asia-Pacific region has emerged as a significant player, with exchanges such as the National Stock Exchange of India (NSE), the Japan Exchange Group (JPX), and the Hong Kong Exchange (HKEX) playing critical roles. In 2023, the Asia-Pacific region experienced the highest growth in trading volume, reaching 103.5 billion contracts, largely driven by India's rapid market expansion. Notably, the NSE alone traded 84.8 billion contracts, reflecting a 123% increase from 2022. Other important markets include Brazil's B3 in Latin America, Eurex and Euronext in Europe, as well as smaller but essential contributors such as Australia's ASX and South Korea's KRX.

The geographical spread of options trading is further illustrated by the figures from 2023: North America accounted for 17.9 billion contracts, while Latin America, Europe, and the Middle East/Africa recorded 8.6 billion, 4.9 billion, and 2.4 billion contracts, respectively. This distribution underscores the strategic significance of regional exchanges in serving both localized and global trading demands.

The dynamic nature of the options market is evident in its trading volumes, with U.S. exchanges maintaining a dominant position. According to CBOE [2] data, the CBOE accounts for 30% of total U.S. volume, followed by NASDAQ with 28%, and NYSE with 17%. MIAX and BOX contribute 16% and 6%, respectively, while MEMX holds a smaller yet expanding

[1] FIA is the leading global trade organization for the futures, options and centrally cleared derivatives markets, with offices in Brussels, London, Singapore and Washington, D.C.

[2] The CBOE is one of the leading options exchanges globally.

share of 3%.

Retail trading has experienced significant growth, now representing approximately 45% of U.S. equity options volume. This trend, initially driven by the pandemic, persists, further fueled by the rising popularity of short-dated options, such as 0DTE (zero days to expiration) contracts. Retail traders also demonstrate a preference for low-priced contracts (below 0.25), which account for 28% of their activity, compared to 25% across all market participants.

Concentration remains another defining characteristic of the market. The top 100 stocks account for 78% of trading activity, with the top 10 alone contributing nearly 50%. Exchange-traded funds (ETFs) play a significant role, comprising 46% of overall options volume. This highlights the market's reliance on liquid and standardized instruments, particularly in index-based strategies.

Equity-related derivatives dominate the options market, encompassing single-stock options, index options, and exchange-traded funds (ETFs). Notable instruments include the SPY (SPDR S&P 500 ETF Trust), QQQ (Invesco QQQ Trust), and the S&P 500 Index (SPX), which consistently rank among the most actively traded. Large-cap stocks such as NVIDIA (NVDA), Tesla (TSLA), and Apple (AAPL) also generate significant trading activity, underscoring their central role in both institutional and retail trading strategies.

In 2023, the National Stock Exchange of India led global equity index options trading, while U.S. markets concentrated on ETFs and individual equities. This emphasis on highly liquid underlying assets highlights their critical role in efficient price discovery and risk management.

Standard call and put options dominate exchange activity due to their straightforward structure and broad applicability. Among these, short-dated options, such as 0DTE contracts, have gained significant popularity, particularly among retail participants, due to their speculative appeal and low-cost entry points. While standard options, as detailed in [1] , remain central to the market, exotic options, such as barrier, Asian, and lookback options, cater to specialized needs in over-the-counter (OTC) markets. These instruments, tailored to specific strategies, are primarily utilized by institutions for customized risk management solutions. However, their decentralized nature limits the availability of public data.

Options linked to commodities, currencies, and interest rates also represent a significant portion of the market. For example, energy options trading volume increased by 34% in 2023, while agricultural options rose by 31%, driven by macroeconomic trends. Meanwhile, interest rate options experienced an 18% growth, reflecting their importance in managing exposure to fluctuating rates.

The diversity of options trading, encompassing both standard and exotic contracts, underscores the market's capacity to address a wide range of financial objectives, from routine hedging to advanced speculative strategies.

Description of the Exotic Options

Asian Options

Asian options were introduced in the 1980s to address challenges in commodity markets such as crude oil and precious metals, which often experience high volatility. Traditional options, tied to a single closing price, can result in unpredictable payouts due to potential price spikes or troughs near expiration. By relying on the average price over time, Asian options provide a more stable valuation basis, reducing sensitivity to extreme price movements and enabling smoother hedging operations. These features make Asian options particularly valuable in markets with frequent price fluctuations. Consequently, they are widely used by corporate treasurers, portfolio managers, and institutional investors to hedge exposures to volatile assets while minimizing the risk of price manipulation near expiration dates.

The payoff of an Asian option depends on the average price of the underlying asset during a predetermined period. For an average price call option , the payoff is: $\max(S_{\text{avg}} - K, 0)$, where S_{avg} is the average of price. The same formula holds for puts. In contrast, for average strike options, the average price serves as the strike price itself. For an average strike call option, the payoff is: $\max(S_T - S_{\text{avg}}, 0)$, where S_T is the price of the underlying at time T. The same principle holds for put options.

The averaging process can be based on either the arithmetic mean or the geometric mean. The arithmetic average is calculated as the sum of observed prices divided by the number of observations or generally called fixings, while the geometric average is the $n\,th$ root of the product of observed prices, assuming n fixings. Arithmetic averages are more sensitive to outliers and tend to yield higher average values in volatile markets. They are more common in real-world applications, particularly in structured products and commodity-linked instruments. Geometric averages, on the other hand, are less sensitive to extreme values and are preferred for mathematical modeling due to their analytical tractability.

From a practical perspective, Asian options are widely employed in sectors where price stability is critical, such as energy, commodities, and foreign exchange markets. For instance, an energy producer might use an Asian call option to hedge against rising fuel costs, ensuring that the average acquisition price remains within budget. Similarly, these options are often integrated into employee stock option plans to mitigate the effects of potential price manipulation around vesting periods.

Barrier Options

Barrier Options are exotic options whose payoff depends on whether the underlying asset's price reaches or exceeds pre-determined level during a set period. Unlike regular options, barrier ones activate or deactivate based on the price hitting the barrier.

First introduced on the OTC market in 1967, barrier options soon gained success among investors, due to their cost efficiency: their pricing is usually lower than vanilla options because the barrier condition reduces the likelihood of the option being exercised, making them more attractive to the market.

Barrier options are classified into two primary types, *knock-out* options and *knock-in* options; each with further subdivisions and different formulas according to the option's nature (call or put).

Knock-in options: only activate if the underlying price hits the barrier during the option's lifetime; otherwise, the option expires worthless. *Knock-out* options: deactivate if the underlying breaches the barrier and remain active only if the barrier is not breached. More specifically, for knock-in options, we distinguish between down-and-in and up-and-in, depending on whether the barrier is breached from the downside or the upside, respectively. Similarly, for knock-out options, we distinguish between down-and-out and up-and-out, depending on whether the barrier is breached from the downside or the upside, respectively.

A key aspect related to the barrier is the fact that they demonstrate distinct sensitivities compared to vanilla options: for knock-out options, higher volatility increases the probability of breaching the barrier, reducing the option's value. This is counterintuitive compared to regular options, where increased volatility increases the option's value. This unique feature makes barrier options effective for hedging strategies and risk management.

Binary Options

Binary options, also known as digital options, are a type of option characterized by discontinuous payoffs. This financial asset can be divided into two different categories: *Cash-or-nothing* and *Asset-or-nothing*.

The first type, in the case of a call option, pays out zero if at maturity T the option price is below the strike price, $S_T < K$, while it pays out a fixed amount Q if the option price is above the strike, $S_T > K$.

The opposite applies to Put options, which pay out zero if the price is above the level of the Strike and pay out Q if it remains below that limit.

On the other hand, an *Asset-or nothing* call does not pay off anything if the underlying price is below the strike price at maturity, $S_T < K$, but it pays off the value of the underlying if it exceeds the strike price, $S_T > K$. About *Asset-or-nothing* put instead, it pays out zero if the underlying value exceeds the strike, while it pays out a sum equal to the underlying if its value is below the strike.

2 Options Pricing Models

In this section, we provide a brief overview of several models used for pricing barrier, binary, and Asian options. Specifically, we consider analytical formulae within the Black-Scholes framework, Monte Carlo methods, and lattice models.

Analytical Formulae and Black Scholes Framework

Analytical formulae for Barrier Options

The pricing of standard barrier options cannot be achieved by directly applying the Black-Scholes-Merton formula due to the distinct conditions imposed by the barrier. Nonetheless, several models have been developed within this theoretical framework. In particular, the models proposed by Merton [27], Reiner and Rubinstein [33], and Rich [31] offer valuable formulae for pricing continuously monitored barrier options.

Down-and-in and Down-and-out Following the notation used in Hull [22], and assuming that H denotes the continuously monitored barrier, the prices for the down-and-in call and down-and-out call options are given by the following expressions:

if $H \leq K$, the price of a down-and-in is:

$$c_{di} = S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(y) - K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N(y - \sigma\sqrt{T}),$$

if $H \geq K$, the price of a down-and-out is:

$$\begin{aligned} c_{do} = {} & S_0 N(x_1) e^{-qT} - K e^{-rT} N(x_1 - \sigma\sqrt{T}) - \\ & - S_0 e^{-qT} \left(\frac{H}{S_0}\right)^{2\lambda} N(y_1) + K e^{-rT} \left(\frac{H}{S_0}\right)^{2\lambda-2} N(y_1 - \sigma\sqrt{T}). \end{aligned}$$

Here, S_0, K, r, and q represent the initial price of the underlying asset, the strike price, the risk-free interest rate, and the dividend yield, respectively. Moreover, $\lambda = \frac{r-q+0.5\sigma^2}{\sigma^2}$, $y = \frac{\ln[H^2/(S_0 K)]}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$, $x_1 = \frac{\ln(S_0/H)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$, $y_1 = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$.
The relationship between a standard call, a down-and-out call, and a down-and-in call is as follows:

$$c = c_{di} + c_{do}.$$

The above formulae refer to cases in which the barrier, and thus the underlying asset, are continuously monitored, representing a limiting case. Instead of continuous monitoring, the underlying is observed at discrete intervals. Broadie, Glasserman, and Kou [7] propose some adjustments. Specifically, following their notation, we define: $V(H)$ as the price of the

continuously monitored barrier option, $V_m(H)$ as the price of the discretely monitored barrier option, and H' as the adjusted barrier price. Thus, we have the following:

$$V_m(H) = V(H') + o\left(\frac{1}{\sqrt{m}}\right).$$

The adjusted barrier price is:

$$H' = He^{\pm\beta\sigma\sqrt{T/m}}.$$

Where: $+$ applies if $H > S_0$, $-$ applies if $H < S_0$, and $\beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} \approx 0.5826$. ζ is the Rienmann zeta function.

Analytical formulae for Binary Options

For binary options, the Black-Scholes framework can be applied more directly. In fact, modifications to this model were proposed by Reiner and Rubinstein [33], as well as by Cox and Rubinstein [12]. We consider again the notation used by Hull [22].

Asset-or-nothing The call price formula for and *Asset-or-nothing* is:

$$c = S_0 e^{-qT} N(d_1).$$

Cash-or-nothing Assuming a cash amount equal to Q, then the call price formula for a *Cash-or-nothing* is:

$$c = Qe^{-rT} N(d_2).$$

Where, $d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$, and $d_2 = d_1 - \sigma\sqrt{T}$.

Analytical formulae for Asian Options

The Asian options discussed in these notes are those based on arithmetic and geometric averages. Specifically, within this category, a distinction is made between the continuous and discrete versions of both geometric and arithmetic average options.

Geometric Continuous Average Originally, Kemna and Vorst [24] proposed a solution for pricing continuous geometric average options by treating them as standard options, with adjustments made to the volatility and the cost-of-carry rate (which includes the interest rate and any additional costs). More specifically, the formula for pricing a call option is:

$$c = S_0 e^{(a-r)T} N(d_1) - Ke^{-rT} N(d_2),$$

where, $d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(a + \frac{\sigma_G^2}{2}\right)T}{\sigma_G\sqrt{T}}$, $d_2 = d_1 - \sigma_G\sqrt{T}$, $a = \frac{1}{2}(r - q - \frac{\sigma^2}{6})$, and $\sigma_G = \frac{\sigma}{\sqrt{3}}$.

Geometric Discrete Average In practice, the Asian options have discrete monitoring of the average, more specifically, the value in the BS framework of a geometric discrete average call option is:

$$c = S_0 e^{(a-r)T} N(d_1) - K e^{-rT} N(d_2),$$

where $d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(a + \frac{\sigma_G^2}{2}\right)T}{\sigma_G \sqrt{T}}$, $d_2 = d_1 - \sigma_G \sqrt{T}$. The difference is based on the following formula for Asian geometric volatility:

$$\sigma_G^2 = \frac{1}{n^2 T}\left[\sum_{i=1}^{n} \sigma_i^2 t_i + 2\sum_{i=1}^{n-1}(n-i)\sigma_i^2 t_i\right]. \tag{1}$$

In equation (1), σ_i is the implied BS global from an option that expires at t_i and t_i is the time to fixing i. The term σ_i is observable in the market, see Haug, Haug, and Margrabe [20] for details. Whereas, $a = \frac{\sigma_G^2}{2} + \frac{1}{n^2 T}\sum_{i=1}^{n}(r - q - \sigma_i^2/2)t_i$, where t_i is the time to each fixing, T is the time to maturity and σ_i is the global BS volatility for an option with expiration t_i.

Arithmetic Continuous Average In the case in which the underlying asset is based on arithmetic average, there is no closed formula. The main reason, is related to the fact that, if the asset is assumed to be log normally distributed, the arithmetic average, will not itself have a log normal distribution. However, it is possible to consider some analytical approximations. We refer to the work of Turnbull and Wakeman [35], as well as Black [2]. We define the first and second moment M_1 and M_2 respectively of the average S, during the option life T. More precisely:

$$M_1 = S_0 \frac{e^{(r-q)T} - 1}{(r-q)T},$$
$$M_2 = \frac{2e^{(2r-2q+\sigma^2)T} S_0^2}{(r-q+\sigma^2)(2r-2q+\sigma^2)T^2} + \frac{2S_0^2}{(r-q)T^2}\left(\frac{1}{2(r-q)+\sigma^2} - \frac{e^{(r-q)T}}{r-q+\sigma^2}\right).$$

By assuming that the average of S during the option life, is log-normally distributed, with the first two moments defined above, and by exploiting the previous formula for geometric average option, we obtain the following call price:

$$c \approx e^{-rT}[M_1 N(d_1) - K N(d_2)],$$

where $d_1 = \frac{\ln\left(\frac{M_1}{K}\right) + \left(\sigma_A^2 \frac{T}{2}\right)}{\sigma_A^2 \sqrt{T}}$, $d_2 = d_1 - \sigma_A \sqrt{T}$, and $\sigma_A^2 = \frac{1}{T}\ln\left(\frac{M_2}{M_1^2}\right)$. Another interesting approximation is provided by Levy [26], and comparison between these approximations are presented in Levy and Turnbull [25].

Arithmetic Discrete Average As mentioned before, the Asian options have a discrete monitoring, and so discrete fixings. Useful approximations for the arithmetic options are derived in Haug, Haug, and Margrabe [20], and in the work by Curran [14] based on the geometric conditioning approach.

Monte Carlo Option Pricing

The Monte Carlo method represents another class of approaches, commonly used for a wide range of problems. Initially proposed by Boyle [3] for option pricing, the Monte Carlo method is capable of valuing various types of European options. Its versatility lies in its ability to model a broad range of stochastic processes, making it highly adaptable.

The underlying idea of Monte Carlo in option pricing, is based on risk neutral valuation principle. More precisely:

$$c = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)]. \tag{2}$$

The formula (2), represent the risk neutral valuation of a standard European call, which is simply the expected discounted value of the option. The expectation in (2) considers the risk neutral probability measure $\mathbb{Q}$. Via Monte Carlo we estimate the expected value in (2).

The first step of Monte Carlo method is based on simulating sample paths or trajectories from a stochastic process, the one assumed is the Geometric Brownian Motion (GBM), which has the following SDE:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t.$$

Under the risk neutral valuation, the SDE has the following form:

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}. \tag{3}$$

Where in equation (3), r is the risk free rate, and $W_t^{\mathbb{Q}}$ is a Wiener process under risk neutral measure $\mathbb{Q}$. The equation (3) has the following solution:

$$S_t = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t^{\mathbb{Q}}\right). \tag{4}$$

The approach involves generating sample paths from (4), and then the option value is determined using equation (2). In the case of non path-dependent European-style options, only the terminal value of the asset, which is the value of the underlying asset at time T, is of interest.

Monte Carlo for Barrier Options

As previously discussed, for European-style options that are not path-dependent, the focus is solely on the terminal value of the underlying asset. In contrast, barrier options are path-dependent, meaning that additional adjustments must be made. Specifically, depending on the type of barrier, the option either ceases to exist or comes into existence depending on whether the asset price touches the barrier. For instance, assuming discrete-time monitoring of the barrier, thus, a number of dates equal to N, and considering a down-and-out call option, the payoff of the option is given by:

$$C_T = \max(S_T - K, 0)\mathbb{1}(\underset{t\in N}{S_t} > H). \tag{5}$$

The above expression holds for all the simulation involved. However, the above adjustment in the payoff, is not the only correction we need to consider. Indeed, as it occur for lattices, the barrier might fall within two prices or nodes, leading to a bias in the option price, or generally the price crosses the barrier in between two monitoring dates. If we simulate (5), one considers some fixed dates in the monitoring process, due to the discretization of the GBM, and observe if the KO condition is satisfied. If we assume that the number of monitoring dates are large, then we tend to obtain results similar to a continuous monitoring. This is generally true, indeed if we consider the monitoring bias [3] of (5), it shrinks to 0 quite slowly, (as $N^{-1/2}$), however, the bias is positive, leading to an overestimation, at least for a *down-and-out* option. Some solutions are available, in particular, since the problem involved as aforementioned is similar to the one faced in lattices, in particular one can consider the solution proposed by: Boyle and Lau [5], Derman et al. [15], Kamrad and Ritchken [23], Ritchken [30], Cheuck and Vorst [9]. Another approach instead is based on considering the possibility that the option might be knocked-out within monitoring dates. In particular, in these notes we consider the adjustments proposed by Baldi, Caramellino and Iovino [1]. This method is based on sharp large deviation approach by Fleming and James [17], Fleming and Souganidis [16]. The aforementioned problem can, in general, also be addressed by incorporating the barrier correction as previously defined by Broadie et al. [7].

Monte Carlo for Binary Options

The pricing of binary options with Monte Carlo, follows similar adjustments in the payoff as observed in barrier, indeed, for instance, for a *cash-or-nothing* call option, which pays a cash amount Q, for all the simulation involved we need to adjust the payoff as follows:

$$C_T = Q\mathbb{1}(S_T > K).$$

Instead, for the *asset-or-nothing* call, we would have a payoff:

$$C_T = S_T\mathbb{1}(S_T > K).$$

The above formulae simply represent the adjustment that must be incorporated into the Monte Carlo simulation when computing the payoff.

Monte Carlo for Asian Options

Given their payoff structure, adjustments are also required for Asian options. Specifically, depending on the type of Asian call option, the payoff must be adjusted as follows for all the simulations involved: considering N dates we have

$$C_T = \max(S_{\text{avg}} - K, 0).$$

[3] The monitoring bias can be seen as the difference between the expected value of the payoff discretely monitored and the expected value of the payoff in case of continuous monitoring.

In this case, S_{avg}, can be the geometric average or the arithmetic average.

Binomial Model

The binomial method is a widely employed numerical technique for pricing options on stocks, futures, and currencies. It was introduced by Cox, Ross, and Rubinstein [13] and Rendleman and Bartter [29]. They defined a recombining binomial tree that approximates geometric Brownian motion. As the number of time steps increases, the binomial model converges to the continuous-time Black-Scholes-Merton formula. This method is particularly valuable for pricing American and exotic options, for which no closed-form solutions exist.

One of the well-known binomial models is the Cox-Ross-Rubinstein model, as presented by Cox, Ross, and Rubinstein [13]. In this model, the asset price can either increase by u with probability p, or decrease by d with probability $1-p$. The probabilities, along with the u and d factors, are chosen to match the first two moments of the stock price distribution. While there are various ways to select these parameters, Cox, Ross, and Rubinstein [13] defined the following parameters: $u = e^{\sigma\sqrt{\Delta t}}$, $d = \frac{1}{u}$, and $p = \frac{e^{r\Delta t}-d}{u-d}$, where p represents the probability of the up factor. These parameters assume a risk-neutral world. A simple graphical representation of the CRR binomial tree is provided in the following Figure.

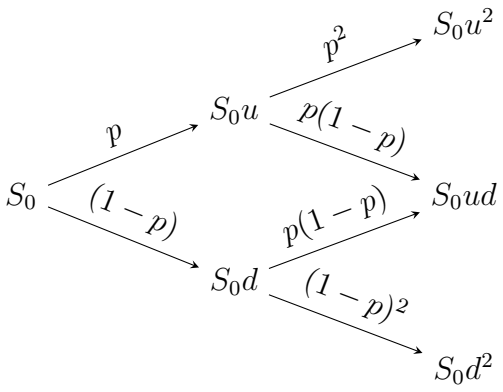


Figure 1: Two step Binomial Tree.

Figure 1 illustrates a two-step binomial tree. This can be generalized to a greater number of steps using the parameters defined above. It is important to note that as the number of steps increases, the binomial model converges to the Black-Scholes formula. The choice of the number of steps typically depends on the type of option being priced, as well as the desired level of accuracy or "tolerance."

The pricing process within the binomial lattice framework is still based on risk-neutral valuation. In general, for European options, we compute the expected value of the payoffs at the terminal nodes, since no early exercise is allowed, under the risk-neutral measure $\mathbb{Q}$, using the probabilities p and $1-p$ defined previously. This expected value is then discounted.

Although, lattices represent a good and simpler alternative, they have some drawbacks. In particular, according to Derman et al. [15], two main sources of error can be identified:

- **Quantization error**: after we choose the number of steps involved, the underlying is allowed to take the values of only those points on the lattice. We are basically considering the underlying moves discretely, therefore the price obtained is correct only for underlying having this quantized behavior.
- **Specification error**: in this case the lattice inaccurately represent the term of the option.

Barrier Options in Binomial Lattices

In general, for barrier options, particularly those used in real-world applications, numerical methods may provide a more suitable alternative.

The pricing procedure for barrier options can resemble that used for standard options. In the case of a knock-out (KO) barrier, the payoff is set to zero at the nodes where the barrier is breached, and the process then proceeds backward, similar to the standard option pricing method. For a knock-in (KI) barrier, the payoff is set to zero if the barrier is not breached, and the process proceeds backward as in the case of standard options, as discussed by Chriss [10].

The approach just mentioned can be problematic, as one of the key issues typically arises when the barrier lies between two nodes. In this case, the lattice may "perceive" a different barrier (referred to as the effective barrier) from the specified one. This issue can be mitigated by increasing the number of steps, which helps reduce the quantization error.

A possible solution is represented by Boyle and Lau [5]. In particular, the following figure enable us to better understand their idea.

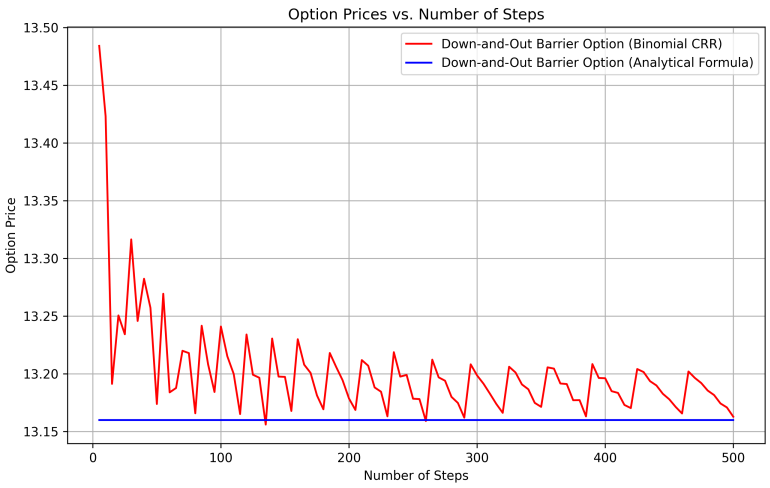


Figure 2: Down-and-Out call price from CRR binomial and BS (Merton) formula vs Number of steps. Figure created by Miura Financial Research using python.

The pattern observed in Figure 2 shows that the option price matches the analytical value only for certain specific steps. This occurs because, at these steps, the barrier aligns almost exactly with the nodes, resulting in zero specification error.

For the CRR tree with N steps, a barrier at stock level B is exactly m nodes away from the current stock price S when $S \exp\left(m\sigma\sqrt{\frac{T}{N}}\right) = B$. In this case, the number of steps can be determined as $N = \frac{m^2\sigma^2 T}{[\ln(S/B)]^2}$. Boyle and Lau present a simple solution that adjusts the number of steps accordingly. This adjustment relies on the fact that the locations of the stock nodes of a CRR tree are independent of the riskless interest rate, and lie at the same stock levels at all times. However, using the Boyle and Lau correction is rarely an ideal approach. In fact, for trees that are not CRR or for time-dependent barriers, the Boyle and Lau correction cannot be applied. An alternative and generally more efficient approach is based on interpolation. Specifically, this method is the one outlined by Derman et al. [15]. This approach typically yields more stable results. The advantages of using interpolation, as compared to a standard binomial approach, are evident. The following figure illustrates this point.

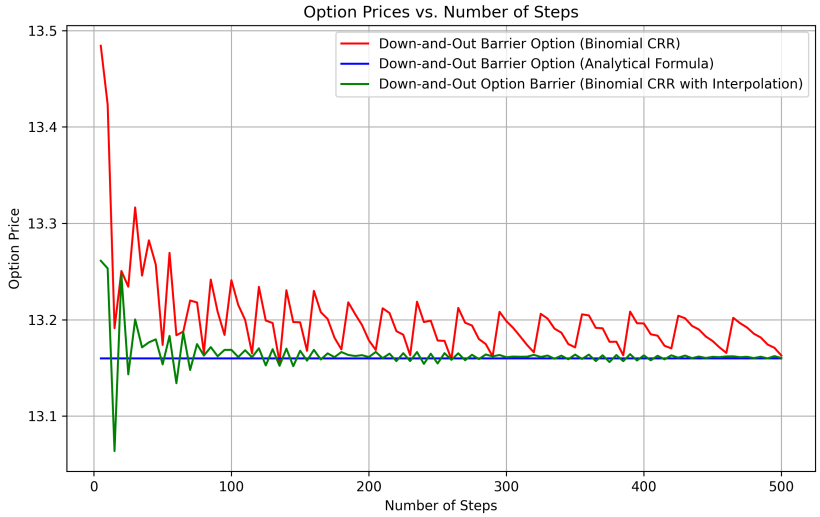


Figure 3: Down-and-Out call price from CRR binomial, CRR binomial with interpolation and BS (Merton) formula vs Number of steps. Figure created by Miura Financial Research using python.

As observed in Figure 3, interpolation significantly improves the results by smoothing the dynamics of the option price and reducing bias. The improvement is evident, with faster convergence compared to the standard approach.

Trinomial Model

Trinomial trees were introduced in option pricing by Boyle [4] and unlike binomial trees, where the asset price can move in only two directions from a given node, trinomial trees allow for three possible directions of movement. As a result, the number of time steps required can be reduced while maintaining the same level of accuracy as a binomial tree. There are multiple methods for selecting the jump sizes and corresponding probabilities in a trinomial tree, all of which yield the same outcome when the number of time steps becomes large. To discretize a geometric Brownian motion, the jump sizes and probabilities must be chosen to match the first two moments of the distribution. One way of choosing the up and down sizes are represented by Haug [21]:

$$u = e^{\sigma\sqrt{2\Delta t}}, \quad d = e^{-\sigma\sqrt{2\Delta t}}, \quad p_u = \left(\frac{e^{r\frac{\Delta t}{2}} - e^{-\sigma\sqrt{\frac{\Delta t}{2}}}}{e^{\sigma\sqrt{\frac{\Delta t}{2}}} - e^{-\sigma\sqrt{\frac{\Delta t}{2}}}} \right)^2$$

$$p_d = \left(\frac{e^{\sigma\sqrt{\frac{\Delta t}{2}}} - e^{r\frac{\Delta t}{2}}}{e^{\sigma\sqrt{\frac{\Delta t}{2}}} - e^{-\sigma\sqrt{\frac{\Delta t}{2}}}} \right)^2, \quad p_m = 1 - p_d - p_u.$$

The parameters defined above are CRR-like. Another way of choosing the parameters of the tree is defined in Hull [22]. A simple graphical representation of the tree defined above, is given in the following Figure:

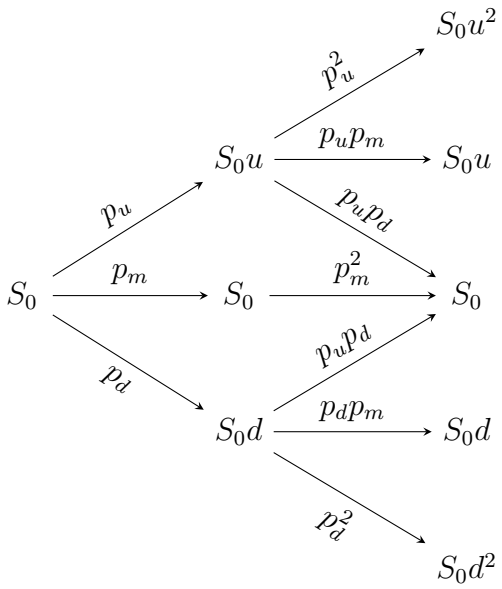


Figure 4: Two step Trinomial Tree.

As shown in Figure 4, the evolution of a trinomial tree is similar to that of a binomial tree, with the key difference being the inclusion of a third step.

The pricing of a trinomial tree follows the same principle as the binomial tree, except that it considers three possible values at each node. In general, for European options, the expected value of the payoffs at the terminal nodes is computed, as early exercise is not permitted. This is done under the risk-neutral measure $\mathbb{Q}$, using the probabilities p_u, p_d, and p_m defined earlier. The expected value is then discounted.

Barrier Options in Trinomial Lattices

As previously discussed, pricing barriers using trees can be challenging; indeed, the same issues generally arise in the trinomial lattice. In this context, apart from using the standard approach—i.e., setting the node that breaches the barrier to zero for knock-out (KO) options and the opposite for knock-in (KI) options—alternative methods are available.

As observed in the binomial model, where the barrier can fall between two nodes, the same problem arises in the trinomial model. An interesting solution is the approach defined by Kamrad and Ritchken [23] and Ritchken [30], commonly referred to as the "stretch" technique.

When this adjustment is applied, the tree parameters are as follows:

$$u = e^{\lambda\sigma\sqrt{\Delta t}} \quad , u = e^{-\lambda\sigma\sqrt{\Delta t}} \quad , p_u = \frac{1}{2\lambda^2} + \frac{\mu\sqrt{\Delta t}}{2\lambda\sigma}, \quad p_m = 1 - \frac{1}{\lambda^2}, \quad p_d = \frac{1}{2\lambda^2} - \frac{\mu\sqrt{\Delta t}}{2\lambda\sigma},$$

where $\mu = r - \frac{\sigma^2}{2}$. The parameter λ is referred to as the "stretch" parameter, which controls the gap between layers of prices on the lattice. When $\lambda = 1$, then $p_m = 0$, and the model reduces to the CRR binomial lattice. Kamrad and Ritchken also describe the procedure for setting the parameter λ such that the barrier is hit exactly. Specifically, assume a down-and-out barrier and assume $\lambda = 1$. We compute the number of down moves that lead to the lowest layer of nodes above the barrier, H. This value, η_0, is the largest integer smaller than η, where η is defined by:

$$\eta = \frac{\ln(S_0/H)}{\sigma\Delta t}.$$

If η is an integer, λ remains unchanged; otherwise, we adjust it by:

$$\eta = \eta_0\lambda.$$

Following this algorithm, $1 \leq \lambda < 2$. This results in a lattice where a layer of nodes coincides exactly with the barrier.

When using the adjustment in the trinomial lattice proposed by Kamrad and Ritchken [23], it is also necessary to consider the adjustment to the barrier, as is common in analytical formulas. In particular, Broadie et al. [6] propose an adjustment to the barrier when dealing with discrete barriers. Specifically, the new barrier is given by:

$$H' = He^{\pm 0.5\lambda\sigma\sqrt{T/m}}.$$

Another important approach for addressing the issues previously highlighted when working with trees is based on interpolation, which follows a similar concept to that employed in the binomial model. This approach is analogous to the one presented by Derman et al. [15]. We can observe the differences between the standard and interpolation approaches, as previously done for the binomial case. The following figure illustrates this point.

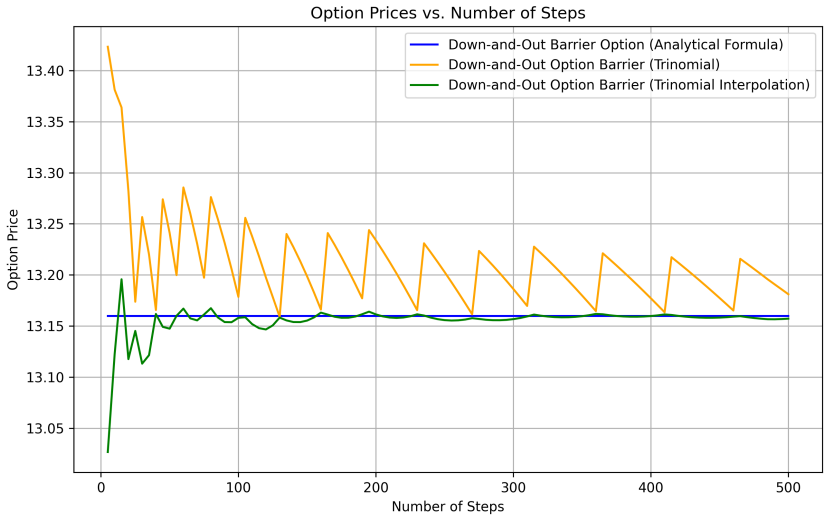


Figure 5: Down-and-Out call price from Trinomial, Trinomial with interpolation and BS (Merton) formula vs Number of steps. Figure created by Miura Financial Research using python.

As seen in Figure 5, the option price dynamics is smoother when interpolation is considered, thereby reducing bias. Additionally, the convergence is faster compared to the standard approach.

3 Numerical Results

In this section, we examine the application of various models for pricing several types of exotic options, specifically: Barrier Knock-Out (KO), Binary, and both Geometric and Arithmetic Asian options. More specifically, we present the results obtained using different methods for pricing these options, based on the models outlined in the preceding sections.

Data and Methodology

For these notes, since all models rely on the calibration or estimation of parameters such as r, the risk-free rate, and σ, which identifies the volatility of the underlying asset, we considered the following data. Specifically, for the risk-free rate r, we used the par yield curve of U.S. Treasuries, selecting a maturity consistent with the option's maturity of one year. For volatility σ, we referred to the S&P 500 index and utilized various historical volatility estimators. These included the sample standard deviation, as well as estimators using OHLC data, namely: Parkinson [28], Garman and Klass [18], Rogers and Satchell [34], and Yang and Zhang [36]. The time period considered for estimation of the parameters goes from 06/12/2023 to 06/12/2024. The data were sourced from Yahoo Finance.

It is important to note that the estimation of σ is typically performed using implied volatility, derived from the market price of options. Similarly, the estimation of the interest rate r is commonly based on OIS (Overnight Indexed Swap) rates, or alternatively, derived from equilibrium relationships using implied volatility and market option prices. Our aim is to provide a straightforward example using real data. While historical volatility and risk-free rates from treasury bills are occasionally used, they are less common in practice. Traders and other market participants typically rely on the aforementioned methods, as these approaches tend to incorporate all available information at the time. For the numerical tests, we consider two scenarios: one in which the option is *in-the-money* and the other in which it is *out-of-the-money*. The strike prices in these cases are generally set at $\pm$ 5% of the initial stock price.

Asian Options Results

In this section, we present the results for both Geometric and Arithmetic Asian options, employing the methods and formulas outlined in the previous sections, specifically those for continuous options. For these types of options, we utilized both the analytical formulae and the Monte Carlo approach. For the Monte Carlo simulations, we conducted 50,000 simulations with a discretization of 252 time steps.

S = 6075 K = 6378 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula	100.69	64.61	65.25	67.80	97.73
Monte Carlo	99.84	64.06	64.69	67.22	96.91
	(0.90)	(0.62)	(0.63)	(0.65)	(0.88)
Monte Carlo Antithetic	99.62	63.87	64.5	67.03	96.7
	(0.55)	(0.39)	(0.39)	(0.40)	(0.53)

Table 1: Arithmetic Asian call option price. Standard error in parenthesis

S = 6075 K = 5771 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula	447.83	428.86	429.15	430.32	446.49
Monte Carlo	446.24	427.48	427.76	428.89	444.5
	(1.67)	(1.34)	(1.35)	(1.37)	(1.64)
Monte Carlo Antithetic	446.1	427.41	427.68	428.81	444.37
	(0.39)	(0.22)	(0.22)	(0.23)	(0.38)

Table 2: Arithmetic Asian call option price. Standard error in parenthesis

S = 6075 K = 6378 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	96.94	62.46	63.07	65.52	94.13
Monte Carlo	97.56	62.83	63.44	65.91	94.73
	(0.87)	(0.61)	(0.61)	(0.63)	(0.85)
Monte Carlo Antithetic	96.76	62.3	62.91	65.36	93.95
	(0.54)	(0.38)	(0.38)	(0.39)	(0.52)

Table 3: Geometric Asian call option price. Standard error in parenthesis

S = 6075 K = 5771 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	440.19	424.18	424.51	425.42	439.43
Monte Carlo	442.91	425.65	425.9	426.94	441.31
	(1.65)	(1.33)	(1.34)	(1.37)	(1.63)
Monte Carlo Antithetic	440.61	423.7	423.94	424.95	439.04
	(0.39)	(0.22)	(0.23)	(0.24)	(0.37)

Table 4: Geometric Asian call option price. Standard error in parenthesis

From Tables 1 and 2, we observe a relatively high precision in the Monte Carlo simulations, with results that are close to the approximation given by the analytical formula in both cases. The standard errors are reduced when employing Antithetic methods, which aligns with the intended objectives of this approach.

In contrast, Tables 3 and 4 show continued good convergence of the Monte Carlo methods to the closed-form solution. These findings generally corroborate existing results and well-established conclusions in the literature.

Barrier Options Results

In this section, we present the results obtained for a *knock-out* call option, specifically focusing on a *down-and-out* option. To evaluate the option's value, we employed three distinct methods: analytical formulas, the Monte Carlo simulation approach, and tree-based techniques. For the Monte Carlo simulations, we incorporated an adjustment method proposed by Baldi, Caramellino, and Iovino [1], which we refer to as the Monte Carlo Probability Adjustment (Monte Carlo Prob adj) in the tables.

In the context of the Monte Carlo simulations, we carried out 50,000 simulations, with each simulation involving a discretization of 252 time steps. The resulting data, showcasing the outcomes of these various methods, are presented in the subsequent tables for comparison.

S = 6075 K = 5771 H = 5400 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	609.91	581.99	582.53	584.70	611.18
Monte Carlo	617.29	587.10	587.62	589.92	615.50
	(2.86)	(2.28)	(2.29)	(2.33)	(2.82)
Monte Carlo Antithetic	620.12	586.87	588.11	590.2	618.05
	(2.84)	(2.25)	(2.22)	(2.2)	(2.80)
Monte Carlo Prob adj	609.81	583.83	583.58	585.72	608.43
	(2.86)	(2.28)	(2.3)	(2.33)	(2.82)

Table 5: Down-and-out call option price. Standard error in parenthesis

S = 6075 K = 6378 H = 5400 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	274.26	208.49	209.73	214.60	269.16
Monte Carlo	276.95	211.25	212.70	217.15	272.27
	(2.03)	(1.53)	(1.53)	(1.57)	(1.99)
Monte Carlo Antithetic	277.23	210.81	211.79	216.55	272.13
	(2.00)	(1.52)	(1.52)	(1.56)	(1.94)
Monte Carlo Prob adj	275.29	209.65	210.84	215.78	270.80
	(2.03)	(1.52)	(1.52)	(1.56)	(1.99)

Table 6: Down-and-out call option price. Standard error in parenthesis

As observed in the Tables above, Tables 5, 6 the basic Monte Carlo method does not demonstrate particularly high efficiency, as indicated by the discrepancy when compared to the Analytical formula. This inefficiency is mainly attributed to the activation of the knock-out condition, as discussed in previous sections, which can occur in various ways, introducing bias into the pricing even with a relatively large number of simulations (e.g., 50,000). One possible remedy is to increase the number of time steps, a strategy also employed in lattice methods. This is because, even in Monte Carlo simulations, the results are highly sensitive to discretization of the sample paths, as noted by Highman [19] and Baldi, Caramellino and Iovino [1], Caramellino and Pacchiarotti [8]. Alternatively, incorporating the correction proposed by Baldi, Caramellino, and Iovino [1], appears to be both particularly effective and less computationally demanding.

Additionally, the results from tree-based models, particularly the Binomial and Trinomial models, were considered. As mentioned in the previous section, the necessary adjustments for

both the binomial and trinomial models were also taken into account. In the following Tables, N, denotes the number of times steps in the tree.

S = 6075 K = 5771 H = 5400 T = $1y$ r = 0.0419 N = 200

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	609.91	581.99	582.53	584.70	607.86
Standard Binomial	613.44	582.59	583.26	586.36	608.83
Interpolation Binomial	609.62	581.95	582.44	584.72	607.67
Standard Trinomial	613.80	583.28	583.93	586.95	609.43
Interpolation Trinomial	609.45	581.90	582.38	584.63	607.60
Ritchken Trinomial	610.05	583.77	584.30	586.73	608.00

Table 7: Down-and-out call option price.

S = 6075 K = 6378 H = 5400 T = $1y$ r = 0.0419 N = 200

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	274.26	208.49	209.73	214.60	269.16
Standard Binomial	274.88	208.65	209.78	214.89	269.30
Interpolation Binomial	274.25	208.60	209.71	214.76	269.08
Standard Trinomial	274.85	208.54	209.64	214.62	269.39
Interpolation Trinomial	273.98	208.44	209.51	214.42	269.94
Ritchken Trinomial	274.38	208.49	209.60	214.64	269.27

Table 8: Down-and-out call option price.

In general, as observed from Tables 5, 6, 7, and 8, tree-based models exhibit superior performance, particularly in approaches that utilize interpolation, which yield strong results. Increasing the number of steps, from 200 to 500 or 1000, would likely have further improved the obtained results. In contrast, Monte Carlo, except for Monte Carlo with prob adj, methods display somewhat lower overall performance. This discrepancy can likely be attributed to the discretization of sample paths, which is especially critical in the context of barrier options. These results largely confirm the general finding that tree-based models tend to outperform the alternative methods presented.

Binary Options Results

We conclude this section by presenting the results for the binary options. Specifically, we consider the following models: Black-Scholes, Monte Carlo, Binomial, and Trinomial models.

The results are provided in the following tables. For the Monte Carlo simulations, we used 50,000 simulations and a discretization of 252 time steps. N denotes the number of steps in the tree. The payout for the *cash-or-nothing* option is denoted by Q.

S = 6075 K = 5771 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252 N = 200

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	4825.89	5165.11	5157.82	5126.09	4848.73
Monte Carlo	4827.71	5165.66	5156.99	5129.45	4850.87
	(12.43)	(10.37)	(10.78)	(10.9)	(12.31)
Monte Carlo Antithetic	4823.25	5162.33	5153.54	5126.79	4846.88
	(5.63)	(5.45)	(5.50)	(5.35)	(5.65)
Standard Binomial	4711.03	5092.90	5089.67	5077.02	4962.87
Standard Trinomial	4879.65	5172.16	5169.11	5157.16	4888.63

Table 9: Asset-or-nothing call option price. Standard error in parenthesis

S = 6075 K = 6378 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252 N = 200

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	3052.20	2977.25	2978.82	2985.03	3046.90
Monte Carlo	3064.03	2988.83	2990.46	2995.86	3058.59
	(15.06)	(14.69)	(14.69)	(14.72)	(15.03)
Monte Carlo Antithetic	3055.45	2980.81	2982.49	2988.17	3050.11
	(4.46)	(4.35)	(4.35)	(4.36)	(5.65)
Standard Binomial	3158.90	3032.44	3027.19	3006.74	3171.84
Standard Trinomial	3166.74	2898.21	2892.97	2872.57	2937.42

Table 10: Asset-or-nothing call option price. Standard error in parenthesis

S = 6075 K = 5771 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252 Q = 1 N = 200

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	0.726	0.793	0.791	0.786	0.731
Monte Carlo	0.73	0.797	0.796	0.789	0.734
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Monte Carlo Antithetic	0.727	0.794	0.793	0.787	0.732
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Standard Binomial	0.707	0.78	0.78	0.777	0.751
Standard Trinomial	0.736	0.794	0.793	0.793	0.738

Table 11: Cash-or-nothing call option price. Standard error in parenthesis

S = 6075 K = 6378 T = $1y$ r = 0.0419 M = $50,000$ Nsteps = 252 Q = 1 N = 200

	Standard	**Parkinson**	**GK**	**RS**	**Yang and Zhang**
Analytical Formula (BS)	0.435	0.434	0.434	0.434	0.435
Monte Carlo	0.438	0.437	0.438	0.437	0.438
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Monte Carlo Antithetic	0.435	0.434	0.434	0.434	0.434
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Standard Binomial	0.452	0.443	0.44	0.437	0.454
Standard Trinomial	0.453	0.422	0.42	0.417	0.418

Table 12: Cash-or-nothing call option price. Standard error in parenthesis

From Tables 9 and 10, we observe that the Monte Carlo methods yield satisfactory results, although the standard errors are notably larger compared to those for other types of options. In contrast, the lattice models generally perform less effectively than the Monte Carlo methods, with the trinomial model typically outperforming the binomial model.

Similar observations apply to Tables 11 and 12, where the results are consistent with those found in the existing literature.

In general, for all the options, we observed varying values depending on the type of historical volatility estimator used. This variation may be attributed to several factors, including gaps or overnight jumps, which are not accounted for by the standard close-to-close volatility. However, the focus of these notes is not to discuss the optimal historical volatility estimators or the calibration of the models employed. Rather, the primary objective is to illustrate the

differences and convergence of the prices across the various models considered.

Conclusions

In conclusion, we summarize the key findings from our analysis of the various types of options considered in this study. In particular, we observe that the estimation of Asian option prices benefits significantly from the use of variance reduction techniques. These techniques lead to a substantial reduction in standard errors, especially when pricing *in-the-money* options. Furthermore, we note that discrepancies in volatility estimates can have a profound impact on the pricing of options, with these effects becoming more pronounced as the strike prices increase and the options transition to being *out-of-the-money*. Consequently, the choice of volatility estimator can have significant implications for the estimated option price, potentially leading to mispricing risks and opportunities.

In the case of barrier options, we employed lattice-based models in addition to the Monte Carlo method for pricing. This dual approach allowed us to address the distinct challenges associated with the discrete nature of lattice-based methods. To mitigate the issues stemming from these methods, we applied interpolation techniques to both binomial and trinomial lattices. The interpolation technique proved to be an effective solution for overcoming the specification and quantization errors that typically impair the precision of price estimates. As a result, the prices derived using this method were notably closer to the analytical values.

For binary options, the estimates obtained through Monte Carlo simulations combined with antithetic variates significantly outperformed the lattice-based models, yielding estimates with more satisfactory standard error magnitudes.

Despite these interesting results, we emphasize that volatility estimation remains a crucial factor in the accurate pricing of these derivatives. The influence of volatility on the option price can often exceed the numerical errors associated with the pricing method itself. Not surprisingly, the estimation of volatility is generally recognized as crucial in options trading. The primary objective of this analysis was to present clear examples of volatility estimation through the use of historical volatility.

In summary, we have observed that when pricing barrier options, it is crucial to carefully consider the methods employed, as certain complications are inherently associated with these specific types of options. This point is particularly evident in the table of results, where the nuances of different methods are highlighted. The complexities in pricing barrier options stem from their unique payoff structure, which introduces additional challenges compared to more traditional option types.

Similarly, the pricing of Asian and binary options remains inherently complex, largely due to the distinctive nature of their payoffs. Despite these complexities, however, these options present opportunities that, when understood and applied correctly, can offer significant advantages for professionals engaged in trading or hedging strategies. The potential for optimization in both risk management and profit generation is considerable, especially for those with a deep

understanding of the associated pricing techniques.

The primary objective of this discussion was to present an overview of the well-established methods commonly used for pricing barrier, Asian, and binary options. In doing so, we also sought to highlight the general challenges that practitioners might encounter when working with these options.

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